THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 10

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 26	$26 \times 26 = 676$
2	**(d)** 22	There are $2n = 22$ numbers between n^2 and $(n+1)^2$
3	**(b)** 10^7	1 crore = 100 lakh = 10,000,000
4	**(b)** XXXVIII	38 = XXXVIII
5	**(b)** parallelogram	Opposite sides are parallel
6	**(a)** 100°	Angle sum of a quadrilateral is 360°
7	**(c)** 3	Digit sum must be a multiple of 9
8	**(a)** 1751	$(100+3) \times 17$ using the distributive property
9	**(b)** 8 : 12 : 15	Make B equal
10	**(c)** division	Ratio $a : b = a/b$
11	**(c)** less than the original	0.99 of original
12	**(d)** ₹ 1800	SP = MP − discount
13	**(a)** $10\sqrt{2}$ cm	Hypotenuse of isosceles right triangle
14	**(b)** No	$10^2 + 11^2 \neq 15^2$
15	**(b)** 2 : 3 : 5	In proportion to investment
16	**(c)** 4	One per base edge
17	**(c)** 30	Total = mean × n
18	**(a)** equal (or as close as possible)	Largest product
19	**(b)** 160 cm²	$\frac{1}{2}(a+b)h$
20	**(b)** doubles	Area ∝ base

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	4.5×10^{-4} (2).	2
22	$8+0+2+1 = 11$ (2).	2
23	Reads the same both ways (1); 121 and 4554 (1).	1+1
24	Shares ₹ 450 and ₹ 750 (2).	2
25	$\sqrt{81+144} = 15$ cm (2).	2
26	1 cm = 2 km (2).	2
27	A few very large incomes distort the mean (2).	2
28	$\frac{1}{2}bh$ (1); bh (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	The digits rise to the number of ones, then fall back (1); $1111^2 = 1234321$ (1); $11111^2 = 123454321$ (1).	1+1+1
30	Only the square (1.5); because a rhombus has unequal diagonals (1.5).	1½+1½
31	$(a^2+2ab+b^2) - (a^2-2ab+b^2) = 4ab$ (3).	3
32	New price 1.2; spending same so quantity $1/1.2 = 83.3\%$ (2); falls by 16.67% (1).	2+1
33	Surface area $= 6 \times 5^2 = 150$ cm² (2); 2 cubes (1).	2+1
34	$(n+3)^2 - n^2 - 9 = 6n$ (2); $6n/n = 6$ (1).	2+1
35	By the theorem, height $= \sqrt{17^2 - 8^2}$ (2); $= 15$ m (1).	2+1
	OR: Triplet $(7, 24, 25)$ (1.5); $49 + 576 = 625 = 625$ (1.5).	1½+1½

36	Trapezium $= \frac{1}{2}(10+26)\times 9 = 162$ cm² (2); rectangle 234 cm² (1).	2+1
	OR: Area $= \frac{1}{2}\times 18\times 12 = 108$ m² (2); cost ₹ 1296 (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 4.8×10^7 (1). (ii) $4.8\times 10^7 + 5\times 10^6 = 5.3\times 10^7$ (2). (iii) 5.3×10^7 (2).	1+2+2
	OR: (i) $\frac{2^8\times 2^3}{2^6} = 2^5 = 32$ (3). (ii) 2^5 (2).	3+2
38	(i) $P\times 1.1664 = 5832$, $P = 5000$ (3). (ii) Interest added to principal each period (2).	3+2
	OR: (i) Percentage of the value of goods or services added to it (2). (ii) ₹ 1890 (3).	2+3
39	(i) Axes, plotted points, trend (3). (ii) Temperature, stock prices (2).	3+2
	OR: (i) Both 74 (2). (ii) A (1). (iii) 8 and 58 (2).	2+1+2
40	(i) 600 m² (2). (ii) Same area (1). (iii) ₹ 4800 (2).	2+1+2
	OR: (i) Half difference 20; height $\sqrt{50^2-20^2}\approx 45.8$ m (2). (ii) $\frac{1}{2}\times 200\times 45.8\approx 4580$ m² (1). (iii) ≈ ₹ 68700 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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